

AKASH KASTURE

Java Backend Engineer

Pune, India | +91 9359346476 | akashkasture0802@gmail.com | linkedin.com/in/akashkasture | github.com/akashkasture

PROFESSIONAL SUMMARY

Java Backend Engineer with **2+ years** of experience building high-throughput distributed systems and RESTful microservices using Java, Spring Boot, Spring Batch, Spring AI, Apache Kafka, Redis, and Oracle. Delivered production systems processing **15,000+ daily messages** at **99.95% uptime**, automated workflows reducing manual effort by **65%**, and built transaction engines handling **\$50M+ monthly volume** with zero errors. Strong in system design, agentic AI application development, database performance tuning, and CI/CD-driven delivery.

TECHNICAL SKILLS

Languages: Java, SQL, PL/SQL

Frameworks: Spring Boot, Spring MVC, Spring Cloud, Spring Batch, Spring Security, Spring AI, Spring Data JPA, Reactive Programming

AI and LLM: Spring AI, Agentic AI, Agentic Workflows, Agent Loop Design, Tool Calling, Advanced RAG, Vector DB, MCP, Guardrails, LLM Evals

Messaging and Caching: Apache Kafka, IBM MQ, Redis

Databases and Storage: Oracle Database, PostgreSQL, pgvector, MinIO (S3-compatible Object Storage)

Cloud and DevOps: AWS, Docker, Kubernetes, Jenkins

Architecture and APIs: Microservices, Event-Driven Architecture, REST APIs, gRPC, API Gateway, Circuit Breaker, Saga Pattern, CQRS, DDD, OAuth 2.0, JWT

WORK EXPERIENCE

Software Engineer | Newgen Software Technology, Pune | January 2024 - Present

Spring Boot, Spring Batch, API Gateway, Microservices, IBM MQ, Oracle, Docker, Jenkins, WebLogic

- Engineered high-volume **Spring Boot microservices** processing **15,000+ daily SWIFT/SFMS messages** (MT103, MT202) across 5,000+ branches via secure MQ-based inbound/outbound pipelines, improving reliability by **30%**, increasing straight-through processing by **30%**, and achieving **99.95% uptime**.
- Optimized end-to-end **Trade Finance workflows** (BG, LC, IRM, ORM), driving a **40% boost in operational efficiency** and **50% reduction in reconciliation mismatches**, and built an automated charge-debit engine processing **\$50M+ monthly** with **zero reconciliation errors**.
- Re-architected a single-threaded amortization posting batch into a partitioned, parallel **Spring Batch pipeline** (1 to 6 concurrent workers) with a self-balancing lock/claim mechanism, reducing runtime from **approximately 5 hours to approximately 1 hour** with crash-safe, exactly-once posting guarantees.
- Tuned high-scale **Oracle workloads** across 10M+ record tables, rewriting non-sargable predicates and adding composite indexes to convert query plans from **TABLE ACCESS FULL to INDEX RANGE SCAN**, and deployed production-grade solutions on WebLogic ensuring high availability under peak load.
- Collaborated in cross-functional **Agile/Scrum** teams, driving code reviews, UAT, and production go-lives. Awarded the **Pinnacle Award** for outstanding contributions to critical releases.

PROJECTS

Payment Gateway - Distributed Payments Processing System

Spring Boot, PostgreSQL, Apache Kafka, Redis, Docker

- Built a **full payment gateway backend** handling the complete merchant checkout lifecycle (order creation, payment capture, refunds, settlements) end-to-end, sustaining **5,000+ concurrent users** at **sub-200ms P95 latency** with zero degradation under load.
- Achieved **exactly-once payment delivery** by combining Kafka transactional producers, PostgreSQL-backed idempotency keys, and Redis deduplication, eliminating **100%** of double-charges and silent event loss even through broker outages and network retries.
- Built secure **merchant integration** with API-key authentication (generation, rotation, revocation with zero-downtime grace period), signed webhook notifications with **3-retry exponential backoff**, and a reconciliation engine computing fees, GST, and net payouts across **1,000+ daily settlements**.

Prompt2App - Agentic Code Generation Platform

Spring Boot, Spring AI, Apache Kafka, MinIO, Redis, PostgreSQL, pgvector, Kubernetes

- Built a full-stack **agentic code generation platform** using Spring AI with an autonomous **plan-execute-reflect agent loop** and tool calling to scaffold, generate, and assemble production-ready applications from natural language prompts in under **30 seconds**, with context preservation across multi-turn edits and **guardrails** to validate generated output before delivery.
- Designed a production-grade **RAG pipeline** using Spring AI + pgvector, embedding and semantically searching project codebases (**4096-dim vectors**) with similarity-threshold filtering and automatic retry-on-miss logic, improving retrieval precision by **40%** by rejecting low-confidence matches instead of always returning top-K.
- Built an autonomous **error-triage agent** implementing a bounded, fail-open reasoning step (LLM-graded verdicts with **8s timeout** and kill switch) on top of a 3-layer noise filter (denylist, Redis dedupe, RAG-grounded diagnosis), cutting unnecessary auto-fix attempts by **70%** while keeping **P99 latency under 3s** via async fallback, validated with Mockito test suite at **100% pass rate**.
- Deployed on **Kubernetes** with Fabric8-based deployment service where each new project workspace spins up an **isolated pod** with live preview, authored manifests for stateful services (pgvector, PostgreSQL) with persistent volumes, and delivered a security-hardening phase (secrets management, CORS, rate limiting) with CI-backed integration tests.

RailReserve - High-Concurrency Train Booking Platform

Spring Boot, Redis, PostgreSQL, Apache Kafka, Docker

- Built an **IRCTC-scale train reservation system** supporting search, general/tatkal booking, waitlist management, and cancellations, handling **10K+ concurrent booking requests per second** with zero double-bookings using distributed locking (Redis), optimistic concurrency control, idempotency keys, and an outbox pattern for guaranteed event delivery via Kafka.
- Implemented a **virtual queue room** for tatkal booking surges with Redis-based rate limiting (**5K requests/sec throttle**), fair-ordering guarantees, and automatic retry with circuit breakers, reducing failed booking attempts by **80%** and improving P99 booking latency from 12s to **under 2s**.
- Designed the platform on **event-driven architecture** with Kafka for async seat allocation, saga-based distributed transactions for payment-booking coordination, and a real-time waitlist promotion engine using Redis atomic counters, achieving **99.9% end-to-end booking consistency** and **3x throughput improvement** over synchronous alternatives.

EDUCATION

Bachelor of Technology (B.Tech) in Electronics and Telecommunication | Pimpri Chinchwad College of Engineering, Pune | 2020 - 2024 | CGPA: 8.6/10